

Reinforcement learning with many discrete actions

GAIPS RL Reading Group
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Introduction

Introduction

Many **real world** problems have large discrete action spaces;

RL has success stories with **small discrete** action spaces;

RL has success stories with **continuous** action spaces;

Large discrete action spaces are troublesome.

Outline

- 1) Background;
- 2) **Action generalization** problem and **solution**;
- 3) The end.

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 - a) Make the action space smaller;
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- 1) Background;
- 2) **Action generalization** problem and **solution**;
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- 5) Future.

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- Laquoudakis and Parr: **Reinforcement learning as classification: leveraging modern classifiers**, 2003.
- He et al.: **Deep reinforcement learning with a natural language action space**, 2015.
- Dulac-Arnold et al.: **Deep reinforcement learning in large discrete action spaces**, 2015.
- Pazis and Parr: **Generalized value functions for large action sets**, 2011.
- Zahavy et al.: **Learn what not to learn: Action elimination with deep reinforcement learning**, 2018.
- Farquhar et al.: **Growing action spaces**, 2020.
- Khetarpal et al.: **What can I do here? A theory of affordances in reinforcement learning**, 2020.
- Majeed and Hutter: **Exact reduction of huge action spaces in general reinforcement learning**, 2020.
- Pierrot et al.: **Factored action spaces in deep reinforcement learning**, 2020.
- Van de Wiele et al.: **Q-learning in enormous action spaces via amortized approximate maximization**, 2020.
- Tennenholtz and Mannor: **The natural language of actions**, 2019.
- Chandak et al.: **Learning action representations for reinforcement learning**, 2019.

Background

Markov decision processes

MDP:

- Is a tuple $(\mathbf{X}, \mathbf{A}, \mathbf{P}, \mathbf{r}, \gamma)$;
- A policy π maps states to actions stochastically;
- The value of a policy is $Q(\pi)$.
- There is an optimal policy π^* with value Q^* ;

Value based methods approximate Q^* and get the greedy π^* ;

Policy based methods directly approximate a **probability distribution** for π^* .

State generalization

In big problems, we need to **generalize** to say things like

- since **this** state is very *similar* to **that** state, then **this** reward must be very similar to **that** reward.

Before we generalize, we need **structure** to say things like

- **this** state is very similar to **that** state.

State generalization

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Learn **from** features.

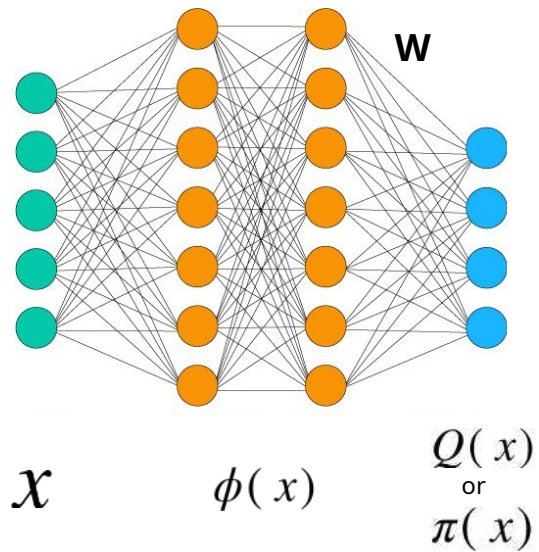
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Learn **the** features.

Reinforcement learning

- discrete actions



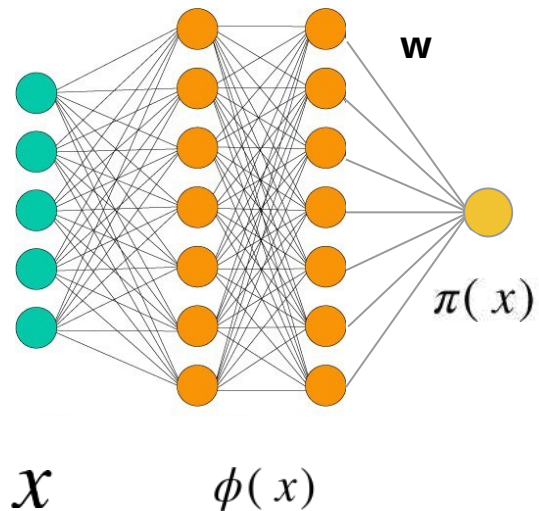
The network learns ϕ that structures the state space;

The network learns a **matrix W** as the final layer;

ϕ allows generalization over the state space.

Reinforcement learning

- continuous actions



The network learns ϕ that structures the state space;

The network learns a **vector w** as the final layer;

ϕ allows generalization over the state space.

1st problem:

Action generalization problem

Action generalization

In problems with many actions, we need to **generalize** to say things like

- since **this** action is very *similar* to **that** action, then **this** reward must be very similar to **that** reward.

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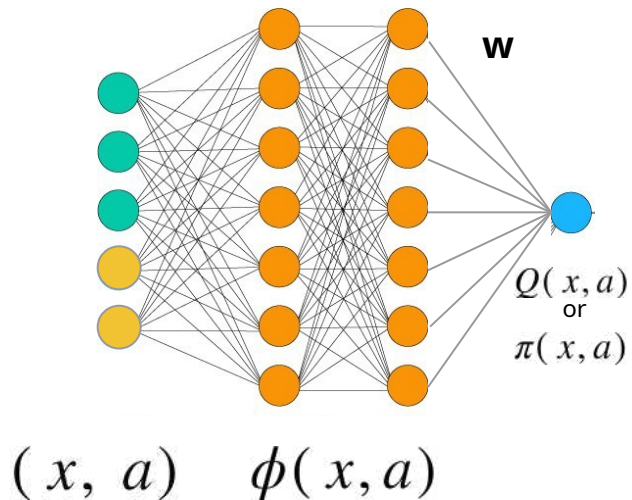
- **this** action is very similar to **that** action.

If only there was a way to learn a structure...

Oh... but there is...



Learning state-action features



The network learns ϕ that structures the state-**action** space;

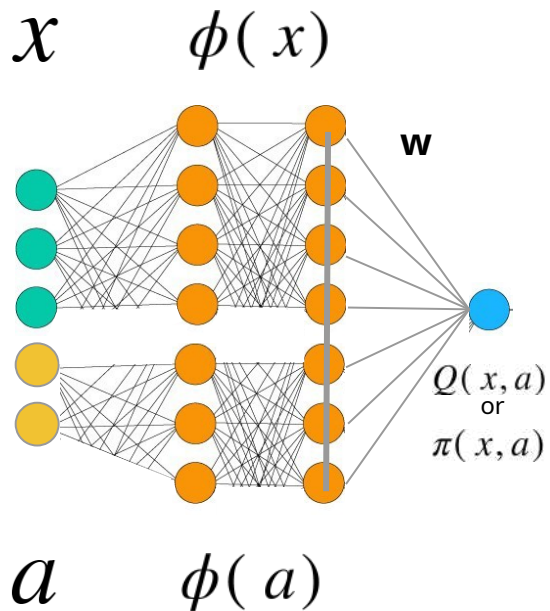
The network learns a **vector** w as the final layer;

ϕ allows generalization over the state-**action** space.

He et al.: **Deep reinforcement learning with a natural language action space**, 2015.

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Learning action-only features



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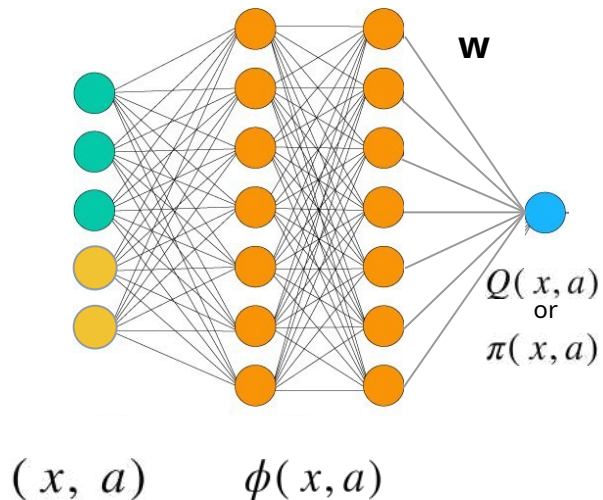
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2nd problem:

Action optimization problem

Action optimization problem



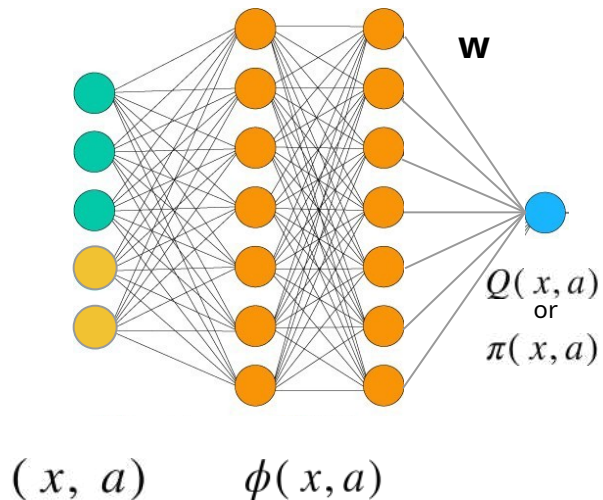
Action computation:

- We observe a state $\mathbf{x}$;
- For each action $\mathbf{a}$, we make one forward pass of $(\mathbf{x}, \mathbf{a})$;
- Then we select the *argmax*.

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Action optimization problem



Action computation:

- We observe a state $\mathbf{x}$;
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- Then we select the *argmax*.

But if we have **thousands** or **millions** of actions...

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Shaping the action space

Reinforcement learning did not suffer from any of the problems in:

1. **small** discrete action spaces;
2. **continuous** action spaces.

We can try to:

1. make the action space **smaller** and still discrete;
2. make the action space bigger but **continuous**.

Reducing the action space

Reducing the action space

Factorize actions.

We can **parallelize** the action computation:

- Consider a decomposition of $\mathbf{A}$ into a n smaller components $\mathbf{A}'$.
 - Output **each** action-component.

If the decomposition is **binary**, we reduce the number of computations exponentially.

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Reducing the action space

Sequentialize actions.

We can **shift** the generalization from the actions to the **states** and the **horizon**:

- Consider a decomposition of $\mathbf{A}$ into n smaller components $\mathbf{A}'$.
 - From $\mathbf{x}$ output $\mathbf{a}[0]$;
 - From $(\mathbf{x}, \mathbf{a}[0])$, output $\mathbf{a}[1]$;
 - ...
 - From $(\mathbf{x}, \mathbf{a}[0], \dots, \mathbf{a}[n-1])$ output $(\mathbf{a}[n])$;

If the decomposition is **binary**, we reduce the number of computations exponentially.

Recurrence can **further** help with optimization.

Majeed and Hutter: **Exact reduction of huge action spaces in general reinforcement learning**, 2020.

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Reducing the action space

Eliminate actions.

We can consider less actions for each decision:

- Eliminate actions permanently and for all states;
- Eliminate actions permanently for each state;
- Eliminate actions for each decision at each state;

We can also start small and **grow** bigger.

Zahavy et al.: **Learn what not to learn: Action elimination with deep reinforcement learning**, 2018.

Vinyals et al.: **Grandmaster level in Starcraft II using multi-agent reinforcement learning**, 2019

Van de Wiele et al.: **Q-learning in enormous action spaces via amortized approximate maximization**, 2020.

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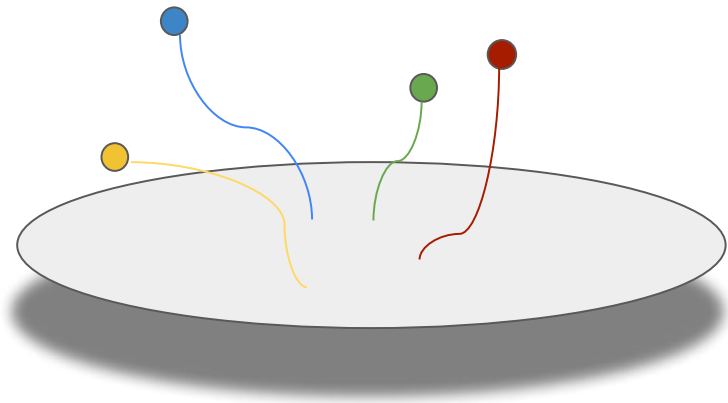
Embedding the action space

Making the action space continuous

Embed actions.

If we **had** action embeddings, we could use policy based methods for continuous action spaces.

Then, we could select the best discrete action with **less** computational complexity.



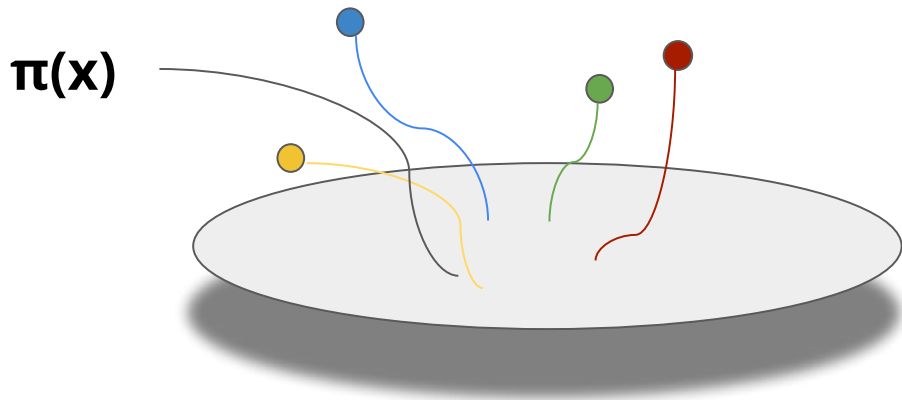
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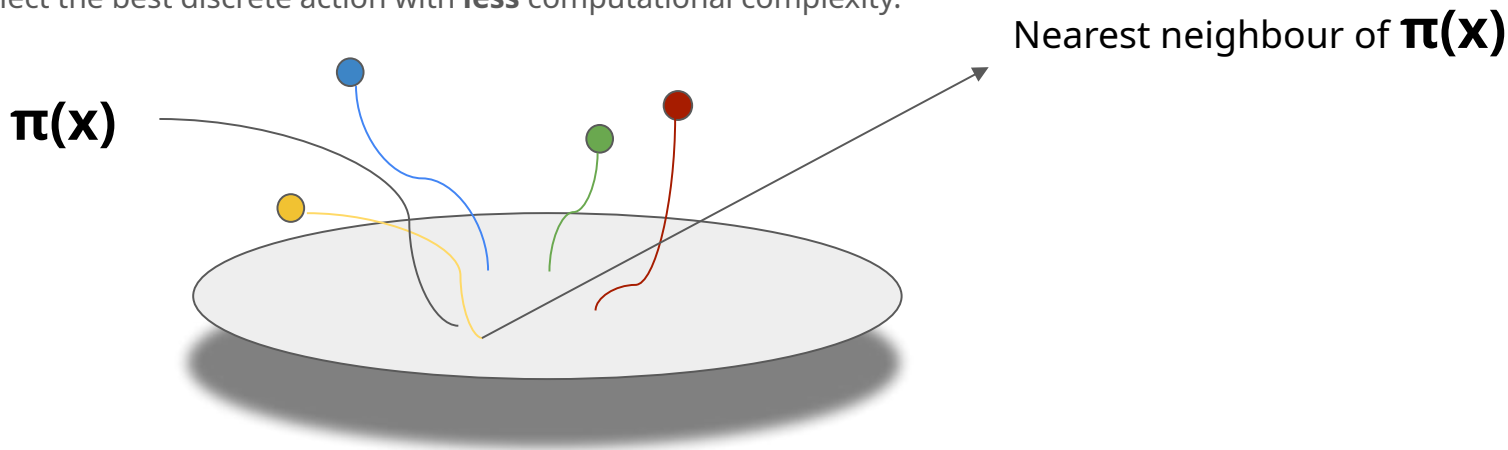


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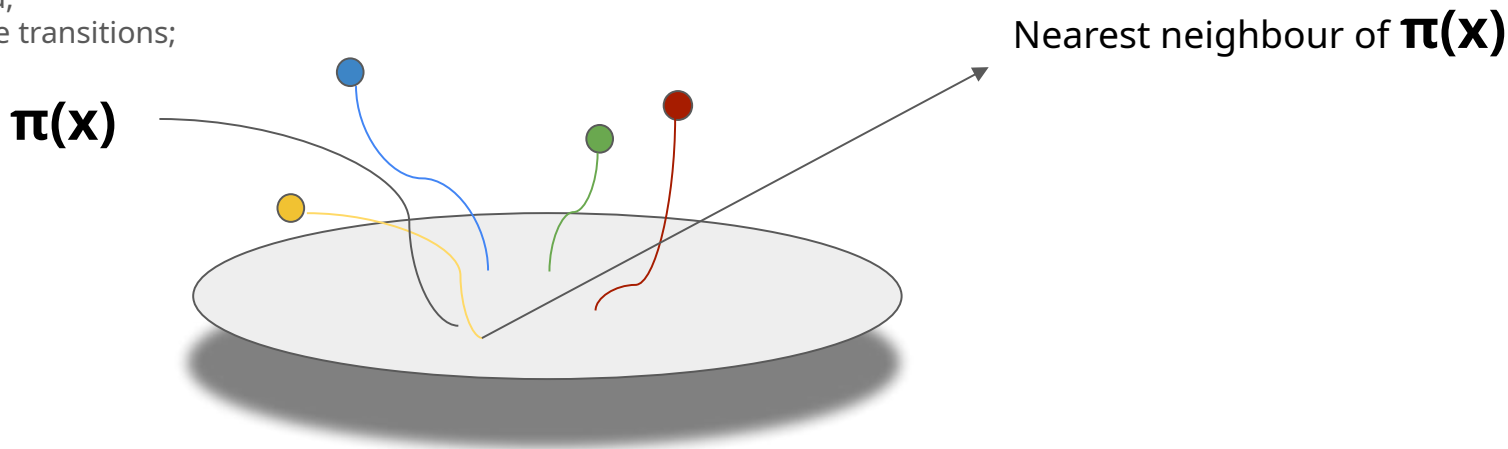


Making the action space continuous

Embed actions.

To **learn** the embeddings:

- Expert data;
- Action state transitions;



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- Learn an intrinsic action elimination signal;
- Set benchmark problems;
- Experimental comparison of methods;
- Understanding which methods can be combined.

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